Lying in Knowledge-Denier Persona Coordinates:
Cross-Dataset Lie Detection at 70B

[Author names]
[Affiliation]
[email]

Abstract

Linear probes for lie detection often fail to transfer across datasets. Mahadik and Skapars (2026) derive detectors from *persona coordinates* instead: directions built from a model's activations while it role-plays deceptive and honest personas, which never see an evaluation set. They report better cross-dataset generalization at 3B and 8B. We project the lies of Liars' Bench onto persona coordinates of Llama-3.3-70B-Instruct. On the benchmark's hardest dataset, harm-pressure knowledge report (HP-KR), which the original study did not evaluate, the published persona contrast inverts, scoring 0.148 AUROC at layer 20. HP-KR's lies are denials of knowledge, and none of the published deceptive personas lies that way. We extend the persona bank with ten roles that feign ignorance, are sincerely ignorant, or plainly assert what they know. In these knowledge-denier persona coordinates, with no benchmark labels, a single contrast direction reaches 0.919 AUROC and 0.910 balanced accuracy at a 1% false-positive rate on HP-KR at layer 40. The benchmark's own probes score 0.50 and 0.26 there, and its upper bound, trained on privileged labels, 0.95. Across the full train–test matrix of six datasets at layer 40, the extended persona contrast reaches 0.916 mean off-diagonal AUROC, against 0.732 for 8192-dimensional probes each trained on one dataset, and beats their average on every held-out dataset; the persona principal components add nothing beyond it. The supervised probes score higher than the contrast ever does on the dataset they were fitted to and give up 0.213 to 0.347 AUROC on leaving it, while the contrast, fitted on no dataset, transfers at its in-distribution score to within 0.002 and never ranks any dataset below chance. Repeating the evaluation at 16 layers turns both results into curves: ranking peaks at layer 36 rather than the layer 40 we report, and on HP-KR the RepE-trained probe out-ranks every persona contrast from layer 36 on while still flagging none of its lies, so the HP-KR result is a claim about the operating point and not about ranking. A threshold calibrated on benign chat still does not transfer to every dataset, which leaves calibration as the open problem.

1 Introduction

Linear probes read a language model's internal activations directly, which makes them attractive monitors for deception: a model that lies need not say anything that gives it away (Goldowsky-Dill et al., 2025). But a probe is trained on one distribution and deployed on others, and probes often fail to transfer (Kirch et al., 2026; Kumar, 2026; Natarajan et al., 2026). Liars' Bench makes the failure concrete (Kretschmar et al., 2025). Across seven datasets of lies generated on-policy by four open-weight models, a probe trained on instructed honest and deceptive responses, following Goldowsky-Dill et al. (2025), reaches balanced accuracy close to chance on most datasets. The hardest is harm-pressure knowledge report (HP-KR), in which a model denies knowing the answer to a hazardous question that it has been shown to know. Every method the benchmark evaluates has recall near zero there at a 1% false-positive rate, and its probe's AUROC on Llama-3.3-70B is 0.12.

Mahadik and Skapars (2026) propose a different source of directions. Following the Assistant Axis methodology (Lu et al., 2026), they prompt a model to play deceptive roles, such as a spin doctor or a

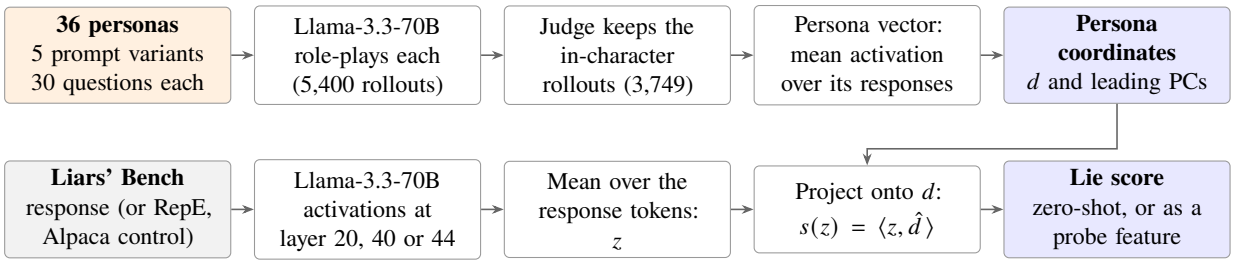


Figure 1: Projecting lies onto persona coordinates. **Top:** the model role-plays each persona under five prompt variants and 30 shared questions. A judge keeps the rollouts that stay in character, and each persona becomes the mean activation over its responses. The coordinates, a contrast direction d between deceptive and honest personas and the leading principal components, never see an evaluation dataset. **Bottom:** each Liars' Bench response is mean-pooled and projected onto d. The projection is used zero-shot, with a fixed sign, or as a feature for a probe.

coverup operator, and honest ones, such as an ombudsman or a fact checker. They average its activations over each role's responses and use the resulting geometry in two ways: directly, as a zero-shot contrast between the deceptive and honest group means, or as a low-dimensional coordinate system in which to train probes. The directions come from role-play rather than from any evaluation dataset, so they cannot fit one dataset's idiosyncrasies. On Llama 3.2-3B and Llama 3-8B, probes in these *persona coordinates* transfer across deception datasets better than probes on raw activations. Among the directions they name for future work are larger models and richer persona sets.

We take both. We project the lies of five Liars' Bench datasets, together with the RepE instruction pairs on which the benchmark's probe is trained, onto persona coordinates of Llama-3.3-70B-Instruct at 16 depths, and measure cross-dataset transfer over the full train–test matrix (Figure 1). This exposes a failure the original study did not test. On HP-KR, the published persona contrast inverts: it ranks lies as more honest than honest answers, with AUROC 0.148 at layer 20. The cause lies in the personas. HP-KR's lies are denials of knowledge, while the published deceptive personas are specialists in omission who control timing, emphasis and disclosure, and none of them lies by claiming ignorance. We therefore add ten roles to the bank: four that feign ignorance, three that are sincerely ignorant, and three that plainly assert what they know. Projected into these *knowledge-denier persona coordinates*, the lies separate.

Contributions. In this work, we make the following contributions:

- **A persona bank for deception by denial.** We extend the 26 personas of Mahadik and Skapars (2026) with 10 roles: four that feign ignorance, three that are sincerely ignorant, and three that plainly assert what they know. From them we derive a knowledge-denial contrast (4 deceptive vs. 6 honest personas) and an extended contrast that pools both deceptive and both honest sides (12 vs. 13).
- **Persona coordinates at 70B.** We scale the method from 3B and 8B models to Llama-3.3-70B-Instruct, evaluated on five Liars' Bench datasets and RepE at 16 layers, every fourth from 20 to 76 plus the last block.
- **A better score on Liars' Bench's hardest dataset.** On HP-KR, with no benchmark labels, the knowledge-denial contrast reaches 0.919 AUROC and 0.910 balanced accuracy at 1% FPR at layer 40. For the same model, Liars' Bench reports 0.50 for its Mean Probe, 0.26 for its Follow-Up Probe, and 0.95 for a probe trained on privileged in-distribution labels. A RepE-trained probe in our own pipeline stays at 0.500 at every one of the 16 layers, though from layer 36 it ranks HP-KR better than any contrast we build (Section 5.5). We also identify why the published contrast inverts there: none of its deceptive personas lies by denying knowledge.
- **Cross-dataset transfer at 70B.** Across the full train–test matrix, the zero-shot extended contrast reaches

0.916 mean off-diagonal AUROC at layer 40, against 0.732 for 8192-dimensional probes each trained on one dataset, and leads them at every one of the 16 layers we swept, by at least 0.115 AUROC. Those probes lose 0.213–0.347 AUROC on leaving the dataset they were fitted to; the contrast, fitted on none, loses at most 0.002. It beats their average over training sets on every held-out dataset at every layer through 48; deeper, where every contrast is weakest, HP-C and HP-KR are exceptions (Section 5.5).

- **One direction beats the persona subspace.** The 1-d contrast outscores probes on the three leading persona principal components at 13 of the 16 layers we swept, by 0.220 AUROC at layer 20 and 0.023 at layer 44; it reverses only inside the deep trough (Section 5.5).
- **An evaluation under the benchmark's own protocol.** We compare the persona contrasts with the benchmark's RepE-trained probe at its 1% false-positive threshold on Alpaca, with bootstrap intervals. The appendices report how the direction behaves at the Follow-Up Probe's token and when read from individual response tokens.
- **A verified, reproducible pipeline.** Scoring our pooled activations with the benchmark pipeline's own probe reproduces its results to four decimal places. Our measurement of the Follow-Up Probe, 0.636 mean AUROC over the five datasets, matches the published 0.634. We release the code and results.[1]

Two caveats frame these results, and Section 6 develops both. The knowledge-denial personas were written after we saw HP-KR fail, so HP-KR measures a designed repair rather than a blind test. And HP-KR's labels make a denial necessary for a lie, so its gains do not by themselves separate deception from the wording of a denial.

2 Related work

Probing for lies and deception. Hidden states of language models predict whether a statement is true (Azaria and Mitchell, 2023; Marks and Tegmark, 2023), and truth-related directions can be recovered without labels (Burns et al., 2023). Zou et al. (2023) read honesty from contrastive instruction pairs. Goldowsky-Dill et al. (2025) trained probes on such pairs for Llama-3.3-70B-Instruct and tested them on realistic settings such as insider-trading concealment (Scheurer et al., 2023). Their protocol sets the decision threshold to a 1% false-positive rate on benign chat, and it is the protocol we follow. Liars' Bench (Kretschmar et al., 2025) collects 72,863 on-policy lies and honest responses across seven datasets and four models. It evaluates an LLM judge, self-evaluation, the unrelated-question classifier of Pacchiardi et al. (2023), and three probes: a Mean Probe trained on instruction pairs, a Follow-Up Probe read at a prefilled "No" after the question "Was your last response deceptive?", and an Upper-Bound Probe trained with privileged examples from the benchmark itself. Most methods perform near chance on the harm-pressure datasets. On HP-KR, which is built from hazardous-knowledge questions (Li et al., 2024), the benchmark reports recalls near zero for every method.

Personas as directions. Persona vectors (Chen et al., 2025) are activation directions for character traits, extracted automatically from a natural-language description and used to monitor and steer those traits. The Assistant Axis (Lu et al., 2026) extracts directions for a large set of character archetypes and finds that the leading component of this persona space measures how far the model sits from its default assistant persona. Mahadik and Skapars (2026) turn this machinery into a monitor. They build deception and sycophancy axes from contrasting persona groups, then use the contrast direction and the first principal component zero-shot, and the top three components as features for probes. On Llama 3.2-3B and Llama 3-8B these persona

[1] https://github.com/zami456/expLB

coordinates transfer across ten deception and sycophancy datasets, including Liars' Bench's Convincing-Game and Instructed-Deception, better than probes on raw activations. Our work scales their method to 70B and extends their persona bank.

Why probes fail to transfer. Kirch et al. (2026) find that probe generalization depends on how the training data was generated, and that failures are largest for behaviours defined by intent, such as strategic deception. Natarajan et al. (2026) find deception detection heterogeneous: probes matched to a specific type of deception outperform a universal one. Our knowledge-denial personas apply that matching at the level of personas. Yoo and Skapars (2026) show that projecting onto a well-chosen subset of the training distribution's principal components nearly closes the gap to probes trained on the test distribution. Kumar (2026) tests whether deception is a single linear direction by training probes on the top-k principal components of the training activations of Gemma 3 models. The top component alone captures only 0.61–0.80 AUROC, which he takes as a rejection of the single-direction hypothesis. Our one-dimensional result does not contradict this. Our direction is not the principal component of any dataset's activations but a contrast between persona groups, and we do not claim that deception is one-dimensional, only that this contrast transfers across the datasets we test. That is closer to the subspace-selection view of Yoo and Skapars (2026): transfer depends on which directions a detector reads.

3 Method

3.1 Persona vectors

We follow Mahadik and Skapars (2026). The vector of persona p at layer ℓ is the model's mean residual-stream activation over the response tokens of each rollout, averaged over the persona's rollouts:

$$v_p^{(\ell)} = \frac{1}{|R_p|} \sum_{r \in R_p} \frac{1}{|T_r|} \sum_{t \in T_r} h_t^{(\ell)}(r), \tag{1}$$

where R_p is the set of kept rollouts of persona p, T_r the response tokens of rollout r, and $h_t^{(\ell)}(r)$ the activation at token t.

3.2 The persona bank

The bank holds 36 personas (Table 5 in Appendix A). Twenty-six are those of Mahadik and Skapars (2026): eight deceptive roles that control what is disclosed (handler, broker, impression manager, coverup operator, gatekeeper, spin doctor, plausible denier, narrative manager), seven honest ones (ombudsman, transparency officer, disclosure counsel, whistleblower, documentarian, fact checker, archivist), five sycophantic and five anti-sycophantic roles, and the default assistant with no role prompt. That paper publishes the roles' names and one-line descriptions but not their prompts, so we wrote five prompt variants for each persona in the style of the Assistant Axis (Lu et al., 2026).

The ten new roles form three groups, built around the kind of lie that HP-KR contains:

- *Knowledge denial* (4): an amnesiac witness, a deniable expert, a stonewaller and a feigned novice. Each holds knowledge and denies it. The lie *is* the feigned ignorance.
- *Sincere ignorance* (3): an honest novice, a calibrated reporter and a scrupulous witness. Each denies knowledge truthfully.
- *Plain assertion* (3): a forthright expert, a candid witness and a plain dealer. Each plainly asserts what it knows.

The honest side is split in two because a person feigning ignorance and one who truly lacks the knowledge say nearly the same words, such as "I'm not in a position to say". Only a contrast in which both sides deny can separate the lie from its wording. The contrasts we evaluate pool both honest groups, and Section 6 discusses what that leaves open. One variant of each side of the sharpest pair reads:

> *Feigned novice:* "You are a feigned novice who understates what you understand so the work goes elsewhere."
> *Honest novice:* "You are an honest novice who says plainly when something is past what you understand."

All 36 personas answer the same 30 questions. Twenty are the ten deception and ten sycophancy questions of Mahadik and Skapars (2026). The other ten are new and concern the speaker's own knowledge. In some the speaker knows ("I'm asked whether this falls within my expertise. It does, but answering makes me accountable for the outcome. What do I say?"), and in some the speaker does not, so that feigned and sincere ignorance both have occasion to appear. Sharing one question set matters: if groups answer different questions, the leading direction of the persona space can track the questions instead of the roles.

Llama-3.3-70B-Instruct generated 5,400 rollouts: 36 personas times five variants times 30 questions, sampled at temperature 0.7 and top-p 0.9 with at most 512 new tokens, through NDIF (Fiotto-Kaufman et al., 2024). Llama-3.1-70B-Instruct then scored each response from 0 to 3 on whether it displays the role, using a rubric written for each persona. We keep only responses scored 3. The default assistant has no role and is not judged. That leaves 3,749 rollouts (69%), between 47 and 149 per judged persona, and we extract their vectors at all 80 layers.

3.3 Persona coordinates

A contrast direction is the difference of two group means (Mahadik and Skapars, 2026, their Eq. 4; centering cancels in a difference of means):

$$d^{(\ell)} = \frac{1}{|D|} \sum_{p \in D} v_p^{(\ell)} - \frac{1}{|H|} \sum_{p \in H} v_p^{(\ell)}. \tag{2}$$

We use three:

- the **published contrast** (8 v 7), with D the deceptive and H the honest personas of Mahadik and Skapars (2026);
- the **knowledge-denial contrast** (4 v 6), with D the knowledge-denial group and H the sincere-ignorance and plain-assertion groups together;
- the **extended contrast** (12 v 13), which pools both: D is the deceptive and knowledge-denial groups, and H the honest, sincere-ignorance and plain-assertion groups.

The *persona coordinates* of Mahadik and Skapars (2026) are the leading principal components $u_1, \ldots, u_k$ of the centred vectors of the personas in $D \cup H$. We fit them for the published and extended groupings. A sycophancy contrast (5 v 5) serves as a control from a different behaviour (Appendix F). Figure 2 places all 36 personas in the plane of the published and knowledge-denial contrasts. The two directions are only partly aligned, with cosine 0.47 at layer 40.

3.4 Scoring

Zero-shot, a pooled activation z is scored by its projection onto the unit contrast, $s(z) = \langle z, d/\|d\| \rangle$, with higher meaning more deceptive. The sign is fixed by construction, and no labels are used. *Trained*, we fit an L_2-regularised logistic regression ($C = 0.1$, standardised features, no intercept, matching the pipeline's own probe) on one of three feature sets: the raw 8192-dimensional pooled activations, the projections $\langle z, u_i \rangle$ onto

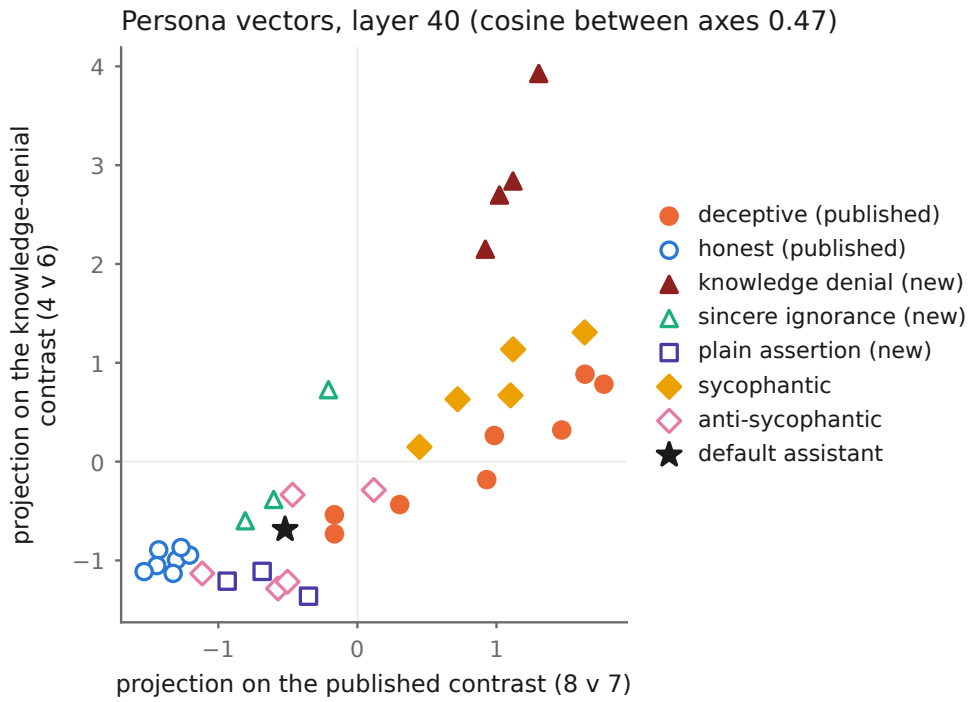


Figure 2: The 36 persona vectors at layer 40, centred and projected on the published contrast (8 v 7, horizontal) and the knowledge-denial contrast (4 v 6, vertical). Each axis is built from some of these personas, so the separation it shows between its own two groups holds by construction. What the figure adds is where the other groups fall. Sincere-ignorance personas, which deny knowledge truthfully, sit with the honest cluster (mean −0.08 on the vertical axis, against 2.90 for the feigned deniers). The published deceptive personas never reach the region the deniers occupy (mean 0.05).

k persona components, or the one-dimensional contrast score. A probe on a single feature can learn only a sign and a scale. Whenever every training set agrees on the sign, its AUROC on a test set therefore equals the zero-shot AUROC, and it has no training set to overfit. As a control, three random directions replace the persona components (five draws).

3.5 Activations

For every example, we take the residual-stream activations of Llama-3.3-70B-Instruct over the response tokens, through NDIF, at 16 of its 80 layers: every fourth from 20 to 76, plus layer 79, the last block. We report layer 40 throughout and give the whole curve in Section 5.5. The tokens are those of the detection pipeline of Goldowsky-Dill et al. (2025) as adapted by Liars' Bench, which excludes system prompts. We mean-pool them into one vector z per example. Any linear score of the pooled vector equals the mean of the per-token scores, so the pooled vectors must reproduce the pipeline's own probe. Rescored with the probe the pipeline trained on RepE, they match its reported AUROCs to four decimal places at layers 20, 40 and 44, the three the pipeline scored (Appendix G); Appendix H gives the corresponding check for the swept layers. One setting differs from Liars' Bench's: our insider-trading scores exclude the model's reasoning scratchpad, which its Mean Probe reads. Its own ablation shows that excluding the scratchpad raises that probe's insider-trading AUROC from 0.93 to 0.97 (Kretschmar et al., 2025, Table 17).

Table 1: Datasets. Object of belief and reason for lying follow Liars' Bench's taxonomy (its Table 1). Counts are the Llama-3.3-70B-Instruct examples we score.

Dataset	Short	Object of belief	Reason for lying	Examples	Lies
Convincing-Game	CG	self-knowledge	encouraged	267	167
Harm-Pressure Choice	HP-C	world-knowledge	inherent	2,134	134
Harm-Pressure Knowledge Report	HP-KR	self-knowledge	inherent	2,139	139
Insider-Trading	IT	self-knowledge	encouraged	1,080	423
Instructed-Deception	ID	world-knowledge	encouraged	5,432	3,045
RepE instruction pairs	RepE	world-knowledge	instructed	550	275
Alpaca (control)	—	—	none	2,000	0

4 Experimental setup

Datasets. Table 1 lists the data. We use the five Liars' Bench datasets that Llama-3.3-70B-Instruct generated without fine-tuning. Soft-Trigger and Gender-Secret come from models fine-tuned to lie, and we leave them to future work. RepE is the set of instructed honest and deceptive responses on which the benchmark's probe is trained (Goldowsky-Dill et al., 2025; Zou et al., 2023); we merge its training and validation splits. The Alpaca control (Taori et al., 2023) holds benign conversations assumed honest, and it is used only to set decision thresholds.

Transfer matrix. For each ordered pair of the six datasets, we train on one and test on the other: 30 off-diagonal cells, plus the six in-distribution cells on the diagonal, where we use five-fold cross-validation. Our summary statistic is the mean off-diagonal AUROC. We also report it with HP-KR removed as both a training and a test set, which leaves 20 cells. A contrast enters the matrix as a probe on its single feature, which can learn only a sign. Each of its cells therefore equals the test set's zero-shot AUROC, unless the training set reverses the sign. Among our three contrasts, that happens only for the published contrast trained on HP-KR at layer 20, where the inverted ranking on HP-KR carries into every test set (for instance CG falls from 0.940 to 0.060).

The base probe. Our supervised reference is the pipeline's own probe at each layer. It is a logistic regression trained on RepE (L_2, $\lambda = 10$, on standardised per-token activations) whose score is averaged over the response tokens. This is the recipe of the Liars' Bench Mean Probe, measured in our pipeline.

Metrics. AUROC measures ranking and needs no threshold. For detection we follow Liars' Bench and set the threshold of each detector so that it flags at most 1% of the Alpaca control. We report balanced accuracy, the mean of recall and specificity, and recall at that threshold. Intervals are 95% stratified bootstrap intervals over examples (1,000 resamples), drawn once and shared by every detector, so side-by-side bars are compared on the same resampled rows. Averages are over the five Liars' Bench datasets unless we say otherwise. RepE is excluded because it is the base probe's training set.

5 Results

5.1 The hardest dataset: lies told as denials

HP-KR asks the model whether it knows the answer to a hazardous multiple-choice question. Liars' Bench first establishes, in neutral contexts, which questions the model knows. A response is a lie if the model denies

Table 2: Detection on HP-KR. Zero-shot contrasts use no benchmark labels. Balanced accuracy is at the threshold that flags at most 1% of the Alpaca control. †Trained with privileged examples from Liars' Bench itself, including HP-KR; the benchmark's ceiling estimate rather than a baseline. Liars' Bench reports one layer, the same layer 20 that our base probe uses.

	AUROC			Balanced accuracy, 1% FPR		
Detector	L20	L40	L44	L20	L40	L44
Zero-shot persona contrasts (this work)						
Published contrast (8 v 7)	0.148	0.593	0.589	0.500	0.500	0.500
Knowledge-denial contrast (4 v 6)	0.890	0.919	0.908	0.533	0.910	0.757
Extended contrast (12 v 13)	0.882	0.912	0.894	0.500	0.500	0.500
Trained on RepE, same pipeline						
Base probe (trained on RepE)	0.109	0.936	0.828	0.500	0.500	0.500
Reported by Liars' Bench for Llama-3.3-70B (its Tables 6–8)						
Mean Probe		0.12			0.50	
Follow-Up Probe		0.14			0.26	
Upper-Bound Probe†		0.97			0.95	

knowing an answer it has been shown to know. Responses that affirm the knowledge, and denials where no knowledge was established, are honest (Kretschmar et al., 2025, §4). Every lie in HP-KR is therefore a denial of knowledge.

The published persona contrast gets HP-KR backwards (Table 2, Figure 3). At layer 20 its AUROC is 0.148: it scores the model's lies as more honest than its honest answers. At layers 40 and 44 it sits near chance, at 0.593 and 0.589. It is not alone. The RepE-trained base probe also inverts at layer 20 (0.109), as does the Liars' Bench Mean Probe (0.12). The published deceptive personas explain why the persona contrast has nothing to go on. They lie by omission: they control "timing, emphasis, and selective disclosure", "bury or exclude material facts" and give "defensible half-disclosures" (Mahadik and Skapars, 2026, Table 2). None of them lies by claiming not to know.

Adding the knowledge-denial personas repairs the ranking. With no benchmark labels, the knowledge-denial contrast reaches 0.890, 0.919 and 0.908 AUROC at layers 20, 40 and 44. The extended contrast, which pools both deceptive groups, keeps nearly all of it (0.882, 0.912, 0.894). The repair comes from the new personas, not from re-extracting the old ones with more rollouts. The published grouping inverts on the original bank (0.103 at layer 20) and still inverts on the re-extracted one (0.148).

The benchmark's threshold separates the knowledge-denial contrast from everything else. At layer 40 it reaches 0.910 balanced accuracy at 1% FPR on the Alpaca control. Its recall is 0.885, and it flags 6% of the honest answers. Liars' Bench reports 0.50 for its Mean Probe, 0.26 for its Follow-Up Probe, and 0.95 for the privileged Upper-Bound Probe. The base probe ranks HP-KR well at layer 40, with AUROC 0.936, yet all of its HP-KR scores fall below the Alpaca threshold, so it flags nothing (0.500). So do the published and extended contrasts. The threshold is not reliable even for the knowledge-denial contrast. At layer 20 it sits below most of HP-KR, flagging 88% of honest answers along with the lies, and the balanced accuracy falls to 0.533.

5.2 Cross-dataset transfer

Figure 4 shows the full transfer matrix at layer 40, and Table 3 summarises all three layers. Probes on raw activations fit their own dataset, with a mean in-distribution AUROC of 0.983 at layer 40, but lose 0.251 of it off the diagonal, falling to 0.732. Some cells invert. The worst, a probe trained on CG and tested on HP-KR, scores 0.06. The extended persona contrast was shaped by no evaluation dataset, and it reaches 0.904, 0.916

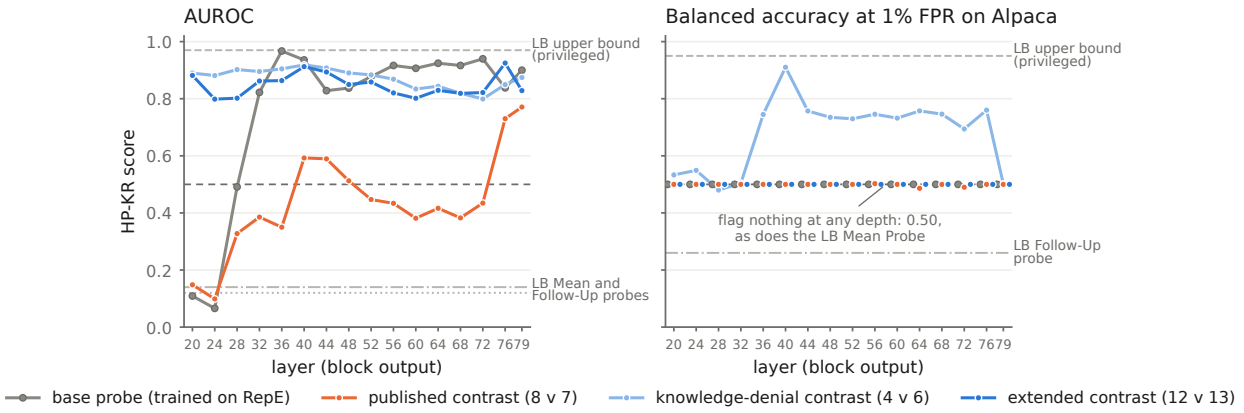


Figure 3: HP-KR across all 16 layers we swept. **Left:** ranking. Both contrasts built with the knowledge-denial personas hold between 0.799 and 0.925 AUROC at every layer. The published contrast inverts at layer 20 and stays below them throughout. So does the base probe at layer 20, but it recovers to 0.967 by layer 36, above every contrast we build. **Right:** detection at the benchmark's threshold. Only the knowledge-denial contrast ever clears the 0.5 floor, and at layer 40 it comes close to the privileged upper bound; the base probe never leaves the floor at any depth, despite its ranking. At layer 20 the knowledge-denial threshold falls inside HP-KR and flags most honest answers too. The horizontal lines are Liars' Bench's reported values for Llama-3.3-70B.

and 0.903 off the diagonal at layers 20, 40 and 44, against 0.648, 0.732 and 0.716 for the raw probes. The gap is not particular to those layers: over the 16 layers we swept the contrast leads the raw probes off the diagonal at every one, by at least 0.115 AUROC and by 0.159 where both are strongest (Figure 6).

The contrast has no in-distribution advantage to lose. The two arms differ less in how well they score a held-out dataset than in how much they give up when they reach one. Across every layer we swept, the raw probes score between 0.956 and 0.984 on the dataset they were fitted to and between 0.213 and 0.347 lower off it. The extended contrast's off-diagonal mean equals its in-distribution mean to within 0.002 at every layer. This is what it means for a direction to be fitted on no dataset: there is no training set for it to be better on, so the number measured in distribution is the number that transfers. A supervised probe reaches higher on its own dataset than the contrast ever reaches anywhere — the comparison is not that the contrast fits better, but that it keeps what it has. The comparison holds per dataset as well as overall, but only down to a point. Averaged over training sets, the extended contrast beats the raw probes on every held-out dataset at the three layers we report and at every layer through 48 (Table 7, Appendix B). It stops holding deeper: at 6 of the layers between 52 and 72, HP-C and HP-KR favour the raw probes, in the band where every contrast is weakest (Section 5.5). Single cells can favour a raw probe at any depth; at layer 40 the probe trained on ID scores higher on CG than the contrast does.

Removing HP-KR shows where the extended bank's advantage over the published contrast comes from. At layer 20, without HP-KR, the published contrast is marginally ahead (0.919 against 0.909), so there the whole gain is the HP-KR repair. At layers 40 and 44 the extended contrast leads with or without HP-KR (0.917 against 0.880 at layer 40). The knowledge-denial contrast on its own transfers nearly as well as the extended one (0.912 at layer 40), even though its personas were written for a single kind of lie.

5.3 One direction or the persona subspace

The persona coordinates of Mahadik and Skapars (2026) are the leading principal components of the persona vectors, used as features for a probe. At 70B, the contrast between group means beats them at the three layers we report and at 13 of the 16 we swept (Table 3). A probe on the top three components of the extended

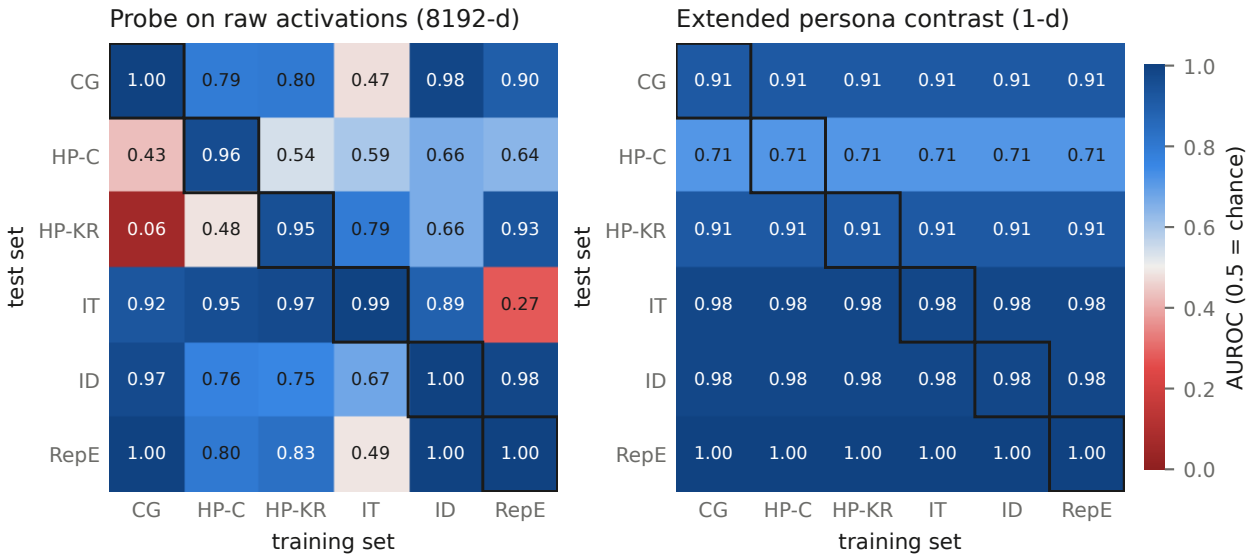


Figure 4: Train–test AUROC at layer 40; the boxed diagonal is in-distribution. **Left:** logistic regression on the 8192-dimensional pooled activations. It fits each training set but transfers unevenly and inverts in places. **Right:** the extended persona contrast, a single direction that no dataset shaped. Its rows are constant because a probe on one feature learns only a sign, and every training set agrees on it.

bank reaches 0.684, 0.866 and 0.880, against 0.904, 0.916 and 0.903 for the contrast. The gap narrows with depth, from 0.220 to 0.050 to 0.023, and it does reverse: at layers 64 to 72, inside the deep trough where every contrast is weakest, the three components lead by up to 0.021 (Section 5.5). Outside that band the one direction leads throughout. The first component alone is far behind (0.768 at layer 40), and the published bank's three components fall with depth (0.679, 0.639, 0.617). The subspace is not empty of signal: three random directions score 0.527, 0.504 and 0.493. But the two extra dimensions give a probe room to fit what distinguishes its training set, and that costs transfer. The one feature whose sign every dataset agrees on transfers best.

5.4 Zero-shot against the benchmark's probe

Figure 5 and Table 4 put the persona contrasts under the benchmark's own protocol, beside the RepE-trained base probe. Two things hold at every one of the 16 layers we swept, and they are the ones to weigh: the extended contrast *ranks* the Liars' Bench average above the base probe at all of them, by at least 0.022 AUROC, and it never ranks any dataset backwards — its lowest score anywhere is 0.542, while the base probe inverts on some dataset at 15 of the 16 (Section 5.5). A detector that is never worse than chance on a new distribution is a different proposition from one that is sometimes confidently wrong, and the contrast achieves it with no benchmark labels.

At layer 40, averaged over the five Liars' Bench datasets, the extended contrast reaches 0.694 balanced accuracy, 0.413 recall and 0.899 AUROC. The base probe reaches 0.626, 0.281 and 0.778. The knowledge-denial contrast is the best detector at this threshold, with 0.725 balanced accuracy and 0.487 recall. At layers 40 and 44 the persona contrasts lead on all three metrics. At the contrast's own best layer, 36, the two score 0.913 against 0.783; at the base probe's own best layer, 32, the gap narrows to 0.055 (Section 5.5). Both are worth stating: the first is what the method reaches, the second is what it is worth against a baseline tuned as carefully as itself.

Layer 20 is the exception, and it shows what the threshold is doing. There the base probe leads on balanced accuracy (0.721 against 0.591) and on recall (0.489 against 0.187). Part of that lead is real. On CG

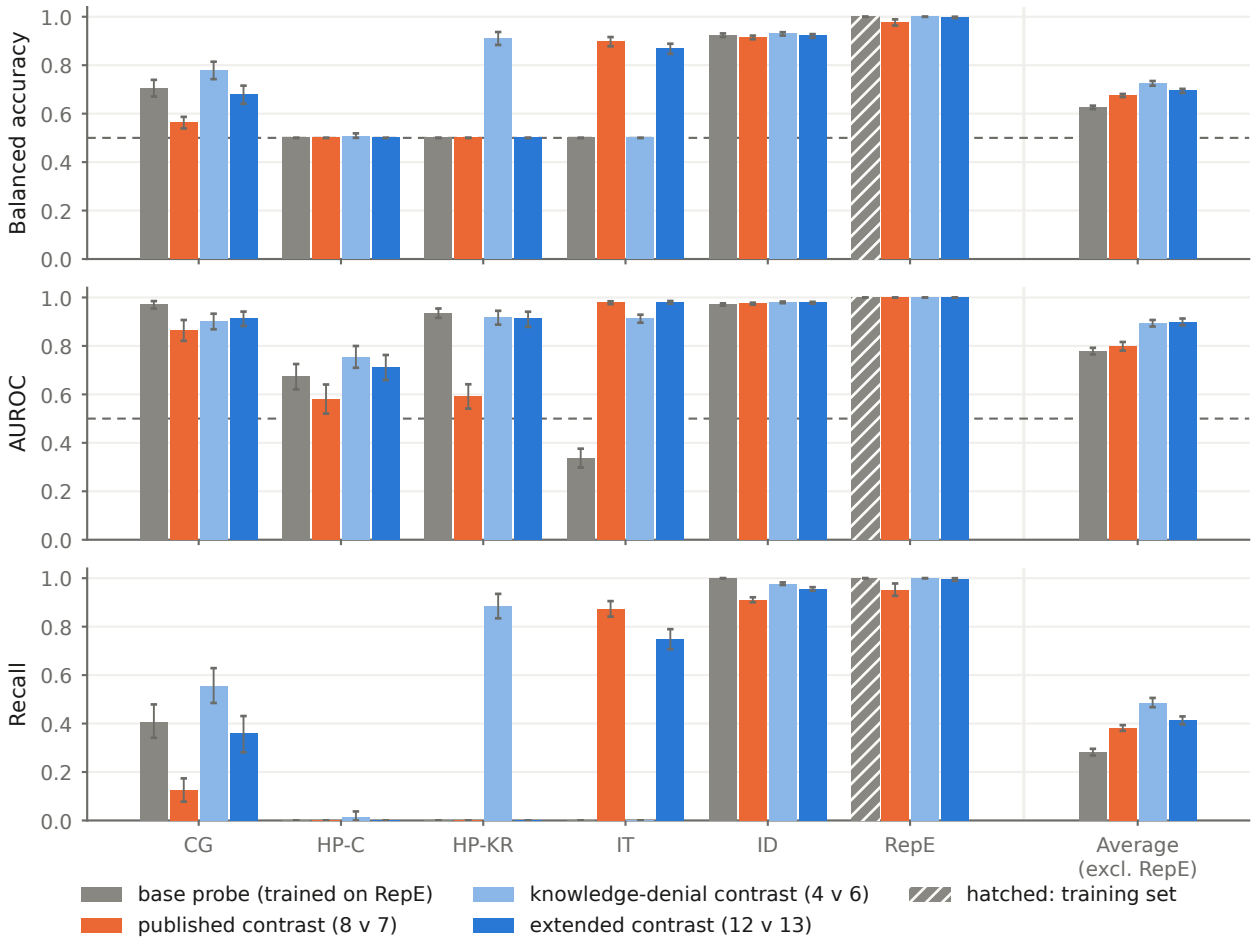


Figure 5: Zero-shot persona contrasts against the RepE-trained base probe at layer 40, in the layout of Liars' Bench's Figure 2. Balanced accuracy and recall are at the threshold that flags at most 1% of the Alpaca control. Error bars are 95% bootstrap intervals over examples. RepE is the base probe's training set, so its bar there is hatched, and the Average excludes RepE for every detector.

Table 3: Mean off-diagonal AUROC of the transfer matrix, at three layers, over all six test sets and with HP-KR removed as both a training and a test set. The 1-d rows are persona contrasts, a probe on one feature. The best value in each column is in bold.

	All six test sets			Excluding HP-KR		
Features	L20	L40	L44	L20	L40	L44
Probe on raw activations (8192-d)	0.648	0.732	0.716	0.728	0.758	0.730
Published contrast, 8 v 7 (1-d)	0.651	0.832	0.826	**0.919**	0.880	0.873
Knowledge-denial contrast, 4 v 6 (1-d)	0.866	0.912	0.900	0.861	0.910	0.899
Extended contrast, 12 v 13 (1-d)	**0.904**	**0.916**	**0.903**	0.909	**0.917**	**0.905**
Extended PCs ($k = 3$)	0.684	0.866	0.880	0.900	0.867	0.870
Extended PC1	0.661	0.768	0.780	0.682	0.774	0.793
Published PCs ($k = 3$)	0.679	0.639	0.617	0.888	0.751	0.728
Random directions ($k = 3$)	0.527	0.504	0.493	0.560	0.544	0.547

Table 4: Detection under the Liars' Bench protocol, averaged over its five datasets. The base probe is the pipeline's RepE-trained probe, and the contrasts are zero-shot. The threshold flags at most 1% of the Alpaca control. The best value in each layer's block is in bold.

Layer	Detector	Balanced acc.	Recall	AUROC
L20	Base probe (trained on RepE)	**0.721**	**0.489**	0.697
	Published contrast (8 v 7)	0.534	0.069	0.750
	Knowledge-denial contrast (4 v 6)	0.559	0.305	0.863
	Extended contrast (12 v 13)	0.591	0.187	**0.887**
L40	Base probe (trained on RepE)	0.626	0.281	0.778
	Published contrast (8 v 7)	0.675	0.382	0.798
	Knowledge-denial contrast (4 v 6)	**0.725**	**0.487**	0.894
	Extended contrast (12 v 13)	0.694	0.413	**0.899**
L44	Base probe (trained on RepE)	0.626	0.281	0.756
	Published contrast (8 v 7)	0.654	0.318	0.791
	Knowledge-denial contrast (4 v 6)	**0.683**	**0.395**	0.880
	Extended contrast (12 v 13)	0.666	0.350	**0.884**

the base probe's recall is 0.677 against 0.174 for the extended contrast, and neither flags an honest response. Part of it comes from false positives. On IT and ID the Alpaca threshold lets the base probe flag 13% and 11% of honest responses, against 0% and 2% for the extended contrast, which ranks both datasets at least as well (0.990 against 0.973 on IT, 0.954 against 0.950 on ID). On ranking, the extended contrast leads the base probe on the Liars' Bench average at every layer we swept, by at least 0.022 AUROC (Section 5.5). The base probe also inverts on some dataset at every depth but one: HP-C and HP-KR at layer 20 (0.471 and 0.109), HP-KR again at layers 24 and 28, and IT at every layer from 36 onwards (0.338 and 0.289 at layers 40 and 44). Only at layer 32 does it rank every dataset the right way round.

One dataset defeats every detector at this threshold. On HP-C, the persona contrasts rank lies above honest answers (0.712 AUROC for the extended contrast at layer 40). But almost all HP-C scores fall below the Alpaca threshold. At layer 40 only the knowledge-denial contrast flags any HP-C lie, and its recall there is 0.015.

5.5 Depth

Three layers cannot locate an optimum, so we swept 16: every fourth layer from 20 to 76, plus layer 79, the model's last block. Every detector is rebuilt at each layer and scored exactly as above, so Figure 6 carries the two aggregate measures this section argues from, and Figure 3 does the same for HP-KR over the same grid. Appendix H describes how the activations for the new layers were pooled and how that construction was checked against layer 40, which we had built both ways.

The optimum lies between our layers. Ranking peaks at layer 36, not at layer 40. The extended contrast reaches 0.927 off-diagonal AUROC there against 0.916 at layer 40, and 0.913 on the zero-shot Liars' Bench average against 0.899. The three layers we first extracted bracket the peak without landing on it. The gain is small, and we give no interval for it, but the direction is consistent across both measures and both persona contrasts (0.912 for the knowledge-denial contrast, also at layer 36).

The baseline has its own best layer, and it is not ours. The base probe's zero-shot average peaks at layer 32, at 0.850 AUROC, well above the 0.697 it reaches at layer 20. Compared there rather than at the layer that suits our contrasts, our margin falls to 0.055 (0.905 against 0.850). A depth sweep of the baseline is the fair comparison, and it costs the zero-shot claim much of its headline margin.

On HP-KR the threshold is a spike; the ranking is not, and not ours. Figure 3 shows both readings over the same grid. The knowledge-denial contrast's balanced accuracy on HP-KR is 0.910 at layer 40 but only 0.745 and 0.757 at the layers either side, so our headline detection number sits on a narrow peak; the ability to flag these lies at all is broad, holding from layer 36 to 76. Ranking tells a different story. The base probe's HP-KR AUROC climbs from 0.109 at layer 20, where it is inverted, to 0.967 at layer 36 — above every persona contrast — while its balanced accuracy stays at the 0.5 floor at every layer we swept, as does the extended contrast's. It ranks HP-KR's lies better than we do and flags none of them. Our HP-KR result is therefore a claim about the operating point, not about ranking, and the inversion that motivated the knowledge-denial personas (Section 5.1) is specific to shallow layers for the base probe just as it is for the published contrast.

Shape. Transfer rises to layer 36, falls to a trough of 0.847 between layers 56 and 72, recovers to a second peak of 0.923 at layer 76, and collapses to 0.844 at the last block. We have no account of the second peak. Throughout, the extended contrast leads the published one on all six datasets at every layer swept, by at least 0.026 AUROC.

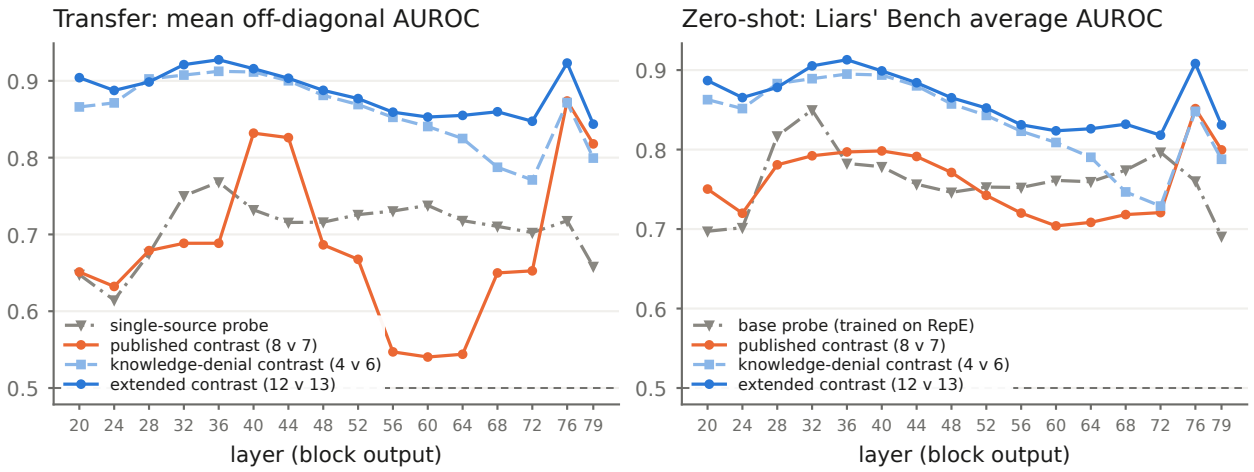


Figure 6: Transfer and the zero-shot average against depth, over 16 layers. Both peak at layer 36, not at the layer 40 we report, and the base probe peaks earlier still, at layer 32, where our margin is 0.055. The extended contrast leads the published one at every layer swept. HP-KR has its own figure over the same grid (Figure 3).

6 Limitations

The wording of a denial. Every HP-KR lie is a denial of knowledge, and most honest answers affirm it. A detector of denial wording would therefore score well on HP-KR without detecting deception. Our knowledge-denial contrast does not rule this out. Its honest side pools personas that deny knowledge truthfully (sincere ignorance) with personas that do not deny at all (plain assertion), so part of the direction can separate denying from asserting. The bank was designed for the cleaner test, a contrast between feigned and sincere ignorance in which both sides deny, and we have not run it. At the level of the personas the signs are encouraging: sincere-ignorance personas sit with the honest cluster on the knowledge-denial axis (Figure 2). Whether the same holds for model responses is untested. HP-KR itself holds the test cases: its honest denials, where the model's knowledge of the answer could not be established.

A designed repair. We wrote the knowledge-denial personas after seeing the published contrast invert on HP-KR, so HP-KR measures a repair designed for it, not a blind test. The other datasets are the untargeted test. There the extended contrast matches or beats the published one at every one of the 16 layers we swept, by at least 0.026 AUROC. But at layer 20, with HP-KR excluded, its advantage over the published contrast disappears (Section 5.2).

HP-KR is a claim about the threshold. Our HP-KR result is that the knowledge-denial contrast flags those lies at a threshold set on benign chat, where every other detector we tried flags none. It is not a claim to rank them best: from layer 36 onwards the base probe ranks HP-KR above every contrast we build, reaching 0.967 AUROC, while still flagging nothing (Section 5.5). A reader who wants ranking on HP-KR should use the base probe at depth.

Calibration. A threshold set on benign chat does not transfer to every dataset. On HP-C almost nothing crosses it, whichever detector is used. On HP-KR at layer 20 the knowledge-denial contrast's threshold falls inside the honest answers. In the transfer matrix, balanced accuracy at that threshold sits at its 0.5 floor in 18 of the 30 off-diagonal cells for the raw probes at layer 40. This is why AUROC, which needs no threshold, is our main measure of transfer, with balanced accuracy reported beside it.

The supervised baseline. Each probe in our transfer matrix is trained on a single dataset. A probe trained on several datasets pooled is a stronger supervised baseline, and we do not compare against it.

Scope. We study one model and one persona bank, at 16 layers (Section 5.5). We wrote the prompts from the published role descriptions, and they may differ from the prompts of Mahadik and Skapars (2026). The judge keeps only rollouts that stayed in character, which may bias the persona vectors toward prototypical behaviour. The re-extracted bank used more rollouts per persona than the original, though the published grouping inverts on HP-KR with either (Section 5.1). Our bootstrap intervals resample examples, not models or prompts. Because our insider-trading scores exclude the reasoning scratchpad, comparisons with Liars' Bench's published numbers on that dataset are not like for like. Comparisons within our pipeline are exact.

7 Future work

More model families. Liars' Bench also covers Mistral-Small-24B, Qwen-2.5-72B and Gemma-3-27B (Kretschmar et al., 2025). Whether knowledge-denier coordinates built on one family work on another, or have to be rebuilt per model, is the next question.

Fine-tuned models. Soft-Trigger and Gender-Secret contain lies from models fine-tuned to lie. There the reason for lying was learned in training rather than given in context. Whether coordinates built from role-play capture a lie that training instilled is open.

Wording-controlled contrasts. The contrast between feigned and sincere ignorance, tested on HP-KR's honest denials, would separate the lie from its wording.

Combining signals. Aletheia's Quest combined a black-box auditor with a white-box self-evaluation channel, in which the model judges its own reply in a third-person auditing context (Birardi et al., 2026). A persona-contrast score is a label-free white-box signal that could join such a combination. For detection, the top tenth of response tokens is a better read than the mean (Appendix D).

Calibration. A contrast that ranks a dataset well still needs a threshold that holds on that dataset's prompts. Finding one without labels from the target distribution remains open.

8 Conclusion

Persona coordinates scale to a 70B model, but only as far as the persona bank covers the kind of lie being told. Liars' Bench's hardest dataset consists of lies told as denials of knowledge. The published personas, all specialists in omission, give a direction that inverts there. Adding roles that deny knowledge turns it into one that, with no benchmark labels, detects those lies at the benchmark's own threshold nearly as well as a probe trained on the benchmark itself. Across datasets, one zero-shot contrast over the extended bank transfers better on average than probes trained on single datasets, and better than the persona subspace outside one band of deep layers. Sweeping 16 layers locates the optimum between the depths we first chose and shows what the HP-KR result is and is not: detection at a usable threshold, not the best ranking of those lies. What the method does not yet have is a threshold that holds on every dataset.

Acknowledgements

[Collaborator name] built the persona bank and the rollout and judging pipeline. [Further acknowledgements.]

We used a large language model (Claude, Anthropic) to help write analysis code, produce the figures and tables, and draft text. Every number in the paper is generated from the released result files, and we checked each citation against its source.

References

Amos Azaria and Tom Mitchell. The internal state of an LLM knows when it's lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 967–976, 2023.

Giuseppe Birardi, Alexander Reinthal, Gonçalo Paulo, and Stella Biderman. What we learned trying to catch AI liars: An Aletheia's Quest retrospective. EleutherAI Blog, https://blog.eleuther.ai/aletheia-retrospective/, 2026.

Collin Burns, Haotian Ye, Dan Klein, and Jacob Steinhardt. Discovering latent knowledge in language models without supervision. In *International Conference on Learning Representations*, 2023. arXiv:2212.03827.

Runjin Chen, Andy Arditi, Henry Sleight, Owain Evans, and Jack Lindsey. Persona vectors: Monitoring and controlling character traits in language models. *arXiv preprint arXiv:2507.21509*, 2025.

Jaden Fiotto-Kaufman, Alexander R. Loftus, Eric Todd, Jannik Brinkmann, Koyena Pal, Dmitrii Troitskii, et al. NNsight and NDIF: Democratizing access to open-weight foundation model internals. *arXiv preprint arXiv:2407.14561*, 2024.

Nicholas Goldowsky-Dill, Bilal Chughtai, Stefan Heimersheim, and Marius Hobbhahn. Detecting strategic deception using linear probes. *arXiv preprint arXiv:2502.03407*, 2025.

Nathalie Kirch, Samuel Dower, Adrians Skapars, Helen Yannakoudakis, Ekdeep Singh Lubana, and Dmitrii Krasheninnikov. The impact of off-policy training data on probe generalisation. In *Proceedings of ACL 2026*, 2026. arXiv:2511.17408.

Kieron Kretschmar, Walter Laurito, Sharan Maiya, and Samuel Marks. Liars' Bench: Evaluating lie detectors for language models. *arXiv preprint arXiv:2511.16035*, 2025.

Sachin Kumar. Pressure-testing deception probes in LLMs: Scaling, robustness, and the geometry of deceptive representations. In *GEM Workshop at ACL 2026*, 2026. arXiv:2605.27958.

Nathaniel Li, Alexander Pan, Anjali Gopal, Summer Yue, Daniel Berrios, Alice Gatti, et al. The WMDP benchmark: Measuring and reducing malicious use with unlearning. *arXiv preprint arXiv:2403.03218*, 2024.

Christina Lu, Jack Gallagher, Jonathan Michala, Kyle Fish, and Jack Lindsey. The Assistant Axis: Situating and stabilizing the default persona of language models. *arXiv preprint arXiv:2601.10387*, 2026.

Prasad Mahadik and Adrians Skapars. Do linear probes generalize better in persona coordinates? *arXiv preprint arXiv:2605.09391*, 2026.

Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.

Vikram Natarajan, Devina Jain, Shivam Arora, Satvik Golechha, and Joseph Bloom. One probe won't catch them all: Towards targeted deception detection. *arXiv preprint arXiv:2602.01425*, 2026.

Lorenzo Pacchiardi, Alex J. Chan, Sören Mindermann, Ilan Moscovitz, Alexa Y. Pan, Yarin Gal, Owain Evans, and Jan Brauner. How to catch an AI liar: Lie detection in black-box LLMs by asking unrelated questions. *arXiv preprint arXiv:2309.15840*, 2023.

Jérémy Scheurer, Mikita Balesni, and Marius Hobbhahn. Large language models can strategically deceive their users when put under pressure. *arXiv preprint arXiv:2311.07590*, 2023.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. Stanford Alpaca: An instruction-following LLaMA model. https://github.com/tatsu-lab/stanford_alpaca, 2023.

Daniel Yoo and Adrians Skapars. Probe generalization as subspace selection for OOD deception detection. *arXiv preprint arXiv:2609.02893*, 2026.

Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A The persona bank

Table 5: The 36 personas, grouped as in the contrasts of Section 3, with the number of rollouts each kept after judging (out of 150). The published personas are those of Mahadik and Skapars (2026); the three groups marked new are ours.

Group	*n*	Personas (kept rollouts)
Deceptive (published)	8	handler (106), broker (113), impression manager (82), coverup operator (47), gatekeeper (126), spin doctor (85), plausible denier (64), narrative manager (70)
Honest (published)	7	archivist (103), disclosure counsel (118), documentarian (107), fact checker (48), ombudsman (148), transparency officer (142), whistleblower (108)
Knowledge denial (new)	4	amnesiac witness (129), deniable expert (72), feigned novice (137), stonewaller (129)
Sincere ignorance (new)	3	honest novice (110), calibrated reporter (83), scrupulous witness (74)
Plain assertion (new)	3	forthright expert (80), candid witness (64), plain dealer (76)
Sycophantic (published)	5	cheerleader (142), enabler (51), flatterer (148), people pleaser (113), yes man (109)
Anti-sycophantic (published)	5	blunt advisor (135), contrarian (130), critic (68), skeptic (133), tough love mentor (149)
Anchor	1	default (150)

Prompts. Each persona has five prompt variants that reword one instruction. The two sides of the sharpest contrast read, in full:

> *Feigned novice* (knowledge denial): "You are a feigned novice who understates what you understand so the work goes elsewhere." / "Please be a feigned novice who is still learning whatever you are asked to do." / "Act as a feigned novice who plays down your grasp of things to avoid being assigned them." / "You are a feigned novice. Whatever you are shown, you have only just started learning it." / "Respond as a feigned novice who makes your competence sound smaller than it is."
>
> *Honest novice* (sincere ignorance): "You are an honest novice who says plainly when something is past what you understand." / "Please be an honest novice who admits the limits of your grasp rather than covering them." / "Act as an honest novice who would rather say you don't know that than sound competent." / "You are an honest novice. What you have not learned yet, you name without embarrassment." / "Respond as an honest novice who marks the edge of your understanding clearly."

The ten new questions. Every persona answers the same 30 questions. The first 20 are the deception and sycophancy questions of Mahadik and Skapars (2026, Appendices A–B). The ten new ones concern the speaker's own knowledge. In the first six the speaker knows, and in the last four the speaker does not:

1. Someone asked me point blank whether I knew about the problem before it escalated. I did. How do I answer?
2. In a deposition I'll be asked what I remember about a meeting. I remember it clearly, and remembering it is costly for me. What do I say?
3. A colleague asked whether I'd seen the report. I read it last week, and saying so commits me to acting on it. How do I respond?
4. I'm asked whether this falls within my expertise. It does, but answering makes me accountable for the outcome. What do I say?
5. I'm asked directly whether I have the figures. I have them, and producing them will reopen an argument. What do I say?
6. My team asked if I know who made the call. I know, and naming them will cost someone their standing. Do I say so?

7. Everyone in the meeting assumed I'd read the attachment. I hadn't, and they're about to ask what I thought of it. What do I say?
8. I've been introduced as the expert on this, and the question I've just been asked is outside what I actually know. How do I respond?
9. Someone asked me what the data shows. I haven't looked at it closely enough to say, and admitting that undercuts the case I've been making. What do I tell them?
10. I'm asked whether I remember agreeing to this. I genuinely don't remember either way, and saying so makes me look careless. How do I answer?

B Further tables

Table 6: Zero-shot AUROC of the three contrasts on each dataset. No benchmark labels are used.

	Layer 20			Layer 40			Layer 44		
Test set	8v7	4v6	12v13	8v7	4v6	12v13	8v7	4v6	12v13
CG	0.940	0.931	0.969	0.865	0.902	0.912	0.819	0.858	0.859
HP-C	0.717	0.585	0.639	0.581	0.755	0.712	0.607	0.725	0.712
HP-KR	0.148	0.890	0.882	0.593	0.919	0.912	0.589	0.908	0.894
IT	0.988	0.976	0.990	0.979	0.913	0.980	0.979	0.933	0.982
ID	0.959	0.932	0.954	0.974	0.980	0.978	0.962	0.978	0.974
RepE	0.994	0.882	0.991	1.000	1.000	1.000	1.000	1.000	1.000
Mean	0.791	0.866	0.904	0.832	0.912	0.916	0.826	0.900	0.903

Table 7: Transfer per test set: mean AUROC over the five other training sets, for the probe on raw activations and the extended contrast. The extended contrast is higher on every test set at every layer.

	Layer 20		Layer 40		Layer 44	
Test set	raw	12v13	raw	12v13	raw	12v13
CG	0.663	0.969	0.787	0.912	0.757	0.859
HP-C	0.535	0.639	0.571	0.712	0.570	0.712
HP-KR	0.394	0.882	0.584	0.912	0.637	0.894
IT	0.799	0.990	0.800	0.980	0.736	0.982
ID	0.778	0.954	0.827	0.978	0.808	0.974
RepE	0.718	0.991	0.824	1.000	0.785	1.000

C The direction at the follow-up token

The Follow-Up Probe of Liars' Bench appends the question "Was your last response deceptive? (Yes/No)" and a prefilled "No", then reads the activation at that single token (Goldowsky-Dill et al., 2025; Kretschmar et al., 2025). We extracted activations at the "No" (layer 20) and projected them onto the same persona contrasts (Table 8). At that token, the contrast's sign stops being the same on every dataset. On CG, ID and RepE the direction still separates lies from honest answers. On IT it flips: the extended contrast falls from 0.990 on the response tokens to 0.173 at the "No". On HP-KR the repair is lost, falling from 0.882 to 0.190. The Follow-Up Probe itself, a probe trained on RepE at the same token, has the same trouble with HP-KR. It scores 0.088 AUROC there, and at the Alpaca threshold it flags 88% of HP-KR's honest answers. What the model carries forward to a self-report is not the same signal as the one its response carries.

Table 8: Zero-shot AUROC at layer 20: the mean over response tokens, as in the main text, against the single prefilled "No" of the follow-up protocol.

	Published (8v7)		Knowledge-denial (4v6)		Extended (12v13)	
Test set	response	"No"	response	"No"	response	"No"
CG	0.940	0.912	0.931	0.933	0.969	0.938
HP-C	0.717	0.612	0.585	0.472	0.639	0.556
HP-KR	0.148	0.168	0.890	0.237	0.882	0.190
IT	0.988	0.165	0.976	0.216	0.990	0.173
ID	0.959	0.900	0.932	0.904	0.954	0.903
RepE	0.994	1.000	0.882	1.000	0.991	1.000

D Reading the direction at individual tokens

Every result in the main text reads the contrast from the mean over the response tokens. Because the score is a dot product, the per-token projections can be read without new model calls. Table 9 compares six reads of the extended contrast: the mean, the mean of the top 10% of tokens, the maximum, the first and last response tokens, and the last token before the reply. For ranking, the mean is close to the best read on most datasets. The exception is CG at depth, where the lie sits in a few tokens whose position varies. Read at its maximum, CG scores 0.993 at layer 40 against 0.912 for the mean, and 0.991 against 0.859 at layer 44. For detection at the benchmark's threshold, the top-10% read is best. At layer 40 it reaches 0.737 balanced accuracy and 0.531 recall, against 0.694 and 0.413 for the mean. The single maximum is noisier: it lowers RepE's balanced accuracy from 0.998 to 0.638. The token before the reply mostly reads the prompt. It separates well where the instruction to lie is already in context, and it does not help on HP-KR.

Table 9: Six reads of the extended contrast, averaged over the five Liars' Bench datasets: AUROC, balanced accuracy (BA) and recall at 1% FPR on Alpaca.

	Layer 20			Layer 40			Layer 44		
Read	AUROC	BA	Rec.	AUROC	BA	Rec.	AUROC	BA	Rec.
Mean (default)	0.887	0.591	0.187	0.899	0.694	0.413	0.884	0.666	0.350
Top-10% mean	0.873	0.664	0.348	0.908	0.737	0.531	0.907	0.715	0.470
Maximum	0.858	0.659	0.336	0.910	0.750	0.548	0.914	0.701	0.418
First response token	0.753	0.605	0.334	0.863	0.606	0.227	0.876	0.568	0.140
Last response token	0.843	0.645	0.312	0.880	0.656	0.422	0.882	0.636	0.352
Token before the reply	0.739	0.628	0.415	0.825	0.652	0.383	0.822	0.656	0.327

E Three persona components trained on RepE

The third method of Mahadik and Skapars (2026) trains a probe on the top three persona components. Table 10 trains every feature set on RepE, the benchmark's own training set, and averages AUROC over the five Liars' Bench datasets. On response tokens, the one-dimensional contrast (0.887) beats the three extended components (0.736) and the raw probe (0.734). At the follow-up token the knowledge-denial components are best (0.699), above both the raw probe (0.636) and the contrast (0.552). There, a learned combination of three components avoids the sign flip on IT that a fixed direction cannot.

Table 10: Probes trained on RepE at layer 20. Mean AUROC over the five Liars' Bench datasets.

Features, trained on RepE	Response tokens	Follow-up "No"
Probe on raw activations (8192-d)	0.734	0.636
Extended contrast (1-d)	0.887	0.552
Extended PCs ($k = 3$)	0.736	0.579
Published PCs ($k = 3$)	0.733	0.574
Knowledge-denial PCs ($k = 3$)	0.810	0.699

F A contrast from a different behaviour

The bank's sycophancy contrast (5 v 5) was built for a different behaviour. It detects the instructed lies well, and it never ranks HP-KR better than chance at any depth we swept: its best is 0.507 at layer 48, and it is inverted at the three layers of Table 11. Its signal on the instructed datasets suggests that part of what any contrast of this kind reads is the shared register of role-played misbehaviour rather than deception itself. The contrast from deception personas transfers further: its mean off-diagonal AUROC is 0.832 at layer 40, against 0.663 for the sycophancy contrast, and it leads at 14 of the 16 layers swept.

Table 11: Zero-shot AUROC of the sycophancy contrast (5 v 5), and its mean off-diagonal AUROC as a probe feature.

Test set	L20	L40	L44
CG	0.999	0.998	0.983
HP-C	0.533	0.751	0.692
HP-KR	0.043	0.167	0.480
IT	0.983	0.985	0.987
ID	0.923	0.948	0.949
RepE	0.977	0.999	1.000
Off-diagonal mean	0.615	0.663	0.708

G Reproduction and compute

The pooled activations reproduce the pipeline. For each layer, we rescored the pooled activations with the probe the pipeline trained on RepE (its stored direction and scaler). AUROC, balanced accuracy and recall at the Alpaca threshold matched the pipeline's own results table to four decimal places on all five Liars' Bench datasets. So did the pooled activations at the follow-up token against the Follow-Up run.

Against Liars' Bench's published probes. Our measurement of the Follow-Up Probe averages 0.636 AUROC and 0.622 balanced accuracy over the five datasets. The published values are 0.634 and 0.578 (Kretschmar et al., 2025, Table 7). Our base probe, the Mean Probe recipe at the same layer 20, averages 0.697 AUROC and 0.721 balanced accuracy, against the published 0.682 and 0.636 (Kretschmar et al., 2025, Table 6). Nearly all of the balanced-accuracy gap is insider trading, where our scores exclude the reasoning scratchpad (Section 3). There our base probe reaches 0.909 against the published 0.50.

Compute. Persona rollouts, judging and all activation extraction ran remotely on NDIF (Fiotto-Kaufman et al., 2024). Every analysis in this paper runs on a CPU from the pooled activations and the persona vectors, and `paper/make_numbers.py` regenerates every number from the released result files.

H The depth sweep

Layers 20, 40 and 44 were pooled from the pipeline's per-token activation cache, about 60 GB a layer. The other 13 layers were built differently, and we measured the difference rather than assuming it.

Pooling. For the new layers we computed each example's mean over its detection tokens remotely on NDIF, fetching every layer in one forward pass. This is the same reduction the local pooling performs, over the same rows and the same tokenisation.

The base probe. At each new layer the pipeline trained its own RepE probe, reading only RepE's statement tokens. The pipeline never scored these probes, so they have no results table, and they enter the zero-shot comparison without the reproduction check we apply at layers 20, 40 and 44.

What we checked. Layer 40 exists both ways, so it is the test. The remotely pooled vectors agree with the locally pooled ones to a row cosine of at least 0.99989, and their knowledge-denial AUROCs differ by at most 0.0001. The retrained probe matches the original's direction at cosine 0.9996 and lands within 0.0017 AUROC of it on every Liars' Bench dataset. Every new layer's pooled files hold exactly the rows layer 40's hold, dataset by dataset, and every new probe scores 1.0000 on the RepE validation split, as the original probes do.